FReCS Net 15 Minute Defence Preparation

基于 2026 年 10 月 28 日编译的最新 108 页论文，供 15 分钟主讲与 5 分钟问答准备。主讲按 13 页设计，总计 14 分 20 秒，留 40 秒给换页、停顿和临场解释。

核心论点：FSDR 有选择地精炼语义与细节，RACE
显式连接描述区域与图像证据；匹配消融显示增益，误差分析说明谁受益。

邮件明确：演讲不能照读；问答不得使用电子辅助。英文段落仅用于练习。正式演讲依靠图和关键词；问答按闭卷口头方式准备，不依赖电脑、手机或备用幻灯片。邮件未附具体评分维度，本材料不假设评分权重。

Slide	Speaking focus	Time window
1	FReCS-Net and the research question	0:00–0:20
2	The task and its two difficulties	0:20–1:20
3	The LViT baseline and the design gap	1:20–2:15
4	Where the modules sit in the network	2:15–3:15
5	FSDR uses semantic context to regulate detail	3:15–4:55
6	RACE compares report cues with image evidence	4:55–6:35
7	A matched four-configuration experiment	6:35–7:35
8	The combined model improves every paired seed	7:35–9:10
9	The gains are concentrated in small infections	9:10–10:50
10	What happens when the description changes	10:50–12:10
11	External context and computational cost	12:10–13:10
12	Limitations and the next evaluation	13:10–13:50
13	Conclusion	13:50–14:20

计时检查点：3:15 开始 FSDR，6:35 进入实验设置，9:10 进入增益来源，12:10 进入外部比较与开销，13:50 开始结论。若落后，优先删去细节公式与外部比较展开，保留主结果、小病灶分析和结论。

Speaking practice and recall cues

先看图和关键词尝试讲述，再用英文参考段落核对信息。无需逐字记忆。

01 FReCS-Net and the research question | 0:00–0:20

Image plus text / infection mask / two contributions

画面建议：论文题目与一个胸片、描述、掩膜示例。标题使用 FReCS-Net。

Good morning. My project is FReCS-Net, a network for segmenting COVID-19 infections in chest X-rays using both an image and a short text description. I will explain two architectural contributions, and show which errors they improve.

讲述提示：开场只说研究对象和目标，不介绍论文目录。

Thesis source: Title page; Abstract; Chapter 1

02 The task and its two difficulties | 0:20–1:20

Faint opacity / coarse location / correct paired description

画面建议：同一病例的胸片、数据集原始描述、GT。最多标出一个模糊病灶边界。

The input is a chest X-ray together with a description of infection location and extent. The output is a binary mask showing the infected pixels. There are two difficulties. First, faint infections overlap with normal anatomy. A sharp change in the image could come from a lesion boundary, but it could also come from ribs or other structures. Second, the text gives coarse cues, such as a side or a lung region, rather than a precise boundary. The model therefore needs to combine regional guidance with visual detail. In this benchmark, these descriptions are supplied at inference as well as during training. They are short structured annotations, rather than unrestricted clinical reports.

讲述提示：解释文本提供大概位置，图像决定边界；明确输入是数据集配套短描述。

Thesis source: Chapter 1; Figure 3.6; Section 4.1

Speaking practice and recall cues

先看图和关键词尝试讲述，再用英文参考段落核对信息。无需逐字记忆。

03 The LViT baseline and the design gap | 1:20–2:15

LViT backbone / pre-added features / regulate detail and report agreement

画面建议：原 PLAM 路径 C+G → PLAM，与分开输入的 FSDR 路径并列。

I build on LViT, which combines a convolutional encoder-decoder with a transformer branch and text features. Its original fusion module, PLAM, receives CNN and reconstructed transformer features after they have already been added together. My first design question is whether keeping these sources separate allows more selective refinement of spatial detail. My second question concerns report guidance: can the network explicitly compare the regions named in the text with evidence in the image before passing skip features to the decoder? These questions lead to FSDR and RACE. Existing work already studies frequency processing and image-text fusion; my contribution is their particular design and controlled evaluation within this backbone.

讲述提示：只讲与你的两个改动直接有关的基线结构，不展开相关工作年表。

Thesis source: Section 3.1; Table 3.2; Chapter 2

04 Where the modules sit in the network | 2:15–3:15

RACE before decoding / FSDR inside decoding / four scales

画面建议：使用最新版论文 Figure 3.1 简化重画，突出 CNN skip → RACE → FSDR decoder；标出未被 RACE 修改的 transformer guide。

This diagram shows the two intervention points. The CNN encoder first extracts features at four spatial scales. RACE refines the skip features that will go to the decoder. The transformer guides and the encoder backbone remain available as separate sources. Each decoder stage then upsamples its current features and applies FSDR using the routed skip, the transformer guide, the decoder features and text. The refined features are concatenated and decoded into the final mask. Both modules operate at four scales, while the additional adaptive frequency filter is used only at the two coarsest decoder stages. This separation lets me test either module on its own and then evaluate their combination.

讲述提示：按箭头走一次：图像编码、RACE 跳连、FSDR 解码、掩膜。不要逐框念名称。

Thesis source: Figure 3.1; Table 3.1

Speaking practice and recall cues

先看图和关键词尝试讲述，再用英文参考段落核对信息。无需逐字记忆。

05 FSDR uses semantic context to regulate detail | 3:15–4:55

Fixed Haar / multi-source support / conditioned detail / residual identity

画面建议：主图只保留低频语义、高频细节和残差三条路径。用 X = Xlow + Xhigh 一条式子说明可重建分解。

FSDR stands for Frequency-aware Semantic-Detail Refinement. Its central idea is to use semantic context to decide which local detail should influence decoding. A fixed Haar transform separates features into low-frequency and high-frequency components. The components reconstruct the original feature, so the decomposition gives an explicit way to process them separately. Low frequency is not automatically foreground, and high frequency is not automatically a lesion edge. The high-frequency path also contains competing texture and artefacts. FSDR combines low-frequency information from the skip, transformer guide and decoder, together with text, to predict channel refinement and spatial support. That support regulates the direct high-frequency component, while local and dilated convolutions learn an additional detail correction. A residual connection retains the original skip. At the two coarsest stages, an adaptive mixture of identity, three-by-three blur and five-by-five blur also refines the decoder and skip features. The design therefore preserves access to detail while controlling how strongly it enters the output. The experiments will show whether this reduces errors.

讲述提示：高频含边界也含纹理和伪影；说明为什么不能全部增强。自适应滤波只用一句话概括。

Thesis source: Sections 3.2.1–3.2.4; Figures 3.2–3.3

06 RACE compares report cues with image evidence | 4:55–6:35

Six coordinate regions / visual evidence / agreement gate / auxiliary training

画面建议：三条关系：text → six region probabilities → prior；CNN → evidence；regional agreement → residual gate。计数头以训练分支淡化。

RACE stands for Report-Anatomy Consistency and Evidence Fusion. It makes the regional relationship between the report and the image explicit. A report head reads the frozen text features and predicts six location probabilities, corresponding to two sides and three vertical levels. A coordinate basis expands these probabilities into a coarse spatial prior. These are image regions, rather than a separate anatomical segmentation. At each encoder scale, another head estimates local visual evidence. Pooling that evidence within the same six regions allows the network to compare visual and textual location estimates. The spatial prior, visual evidence and their agreement jointly gate a learned residual update to the skip. A bounded scale retains the original feature path and controls the size of the correction. During training, parsed descriptions supervise report attributes, and mask-derived regional targets supervise visual evidence. A count head provides an additional training signal but does not directly enter the spatial gate. At inference, RACE uses the image, text and coordinate basis. It does not need the reference mask. I evaluate this routing and its auxiliary training together as the complete RACE module.

讲述提示：六区域是粗坐标分区，不是精确肺叶分割。GT 仅用于训练，不进入推理。

Thesis source: Sections 3.3.1–3.3.4; Figures 3.4–3.5

Speaking practice and recall cues

先看图和关键词尝试讲述，再用英文参考段落核对信息。无需逐字记忆。

07 A matched four-configuration experiment | 6:35–7:35

Four configurations / paired seeds / shared recipe / held checkpoint

画面建议：2×2 开关矩阵；旁边仅列 3 seeds、80 epochs、2,113 Test、Best by Val。

To measure the architectural changes, I use four configurations: the matched PLAM baseline, FSDR, PLAM with RACE, and the combined model. Each is trained with seeds 1219, 2027 and 3407 under the same eighty-epoch single-cycle cosine protocol. The image size, loss, batch size and frozen CXR-BERT encoder are shared. The best checkpoint is selected by validation macro IoU, then evaluated on the same 2,113 test images using a fixed probability threshold above one half. IoU and Dice are averaged per image. This creates same-seed comparisons between configurations and separates module effects from changes in the training recipe.

讲述提示：讲清比较如何成立。主讲统一使用 single cosine，warm restarts 只留作口头问答知识。

Thesis source: Table 4.3; Table 5.2; Section 4.7

08 The combined model improves every paired seed | 7:35–9:10

75.48 to 76.23 / plus 0.75 pp / positive in three seeds

画面建议：四行结果表保留两位小数；配三种子组合增益 0.68、0.79、0.78 pp 的简图。

The combined model achieves 76.23 percent test IoU and 84.71 percent Dice. Compared with the matched PLAM baseline, that is an improvement of 0.75 IoU percentage points and 0.69 Dice points. FSDR alone improves IoU by 0.51 points. RACE improves PLAM by 0.22 points, and adds another 0.24 points when FSDR is already present. The paired-seed plot is important because it shows that the combined gain is positive in every run: approximately 0.68, 0.79 and 0.78 points. Both modules also have positive conditional gains across these three seeds. The combined model has the smallest observed IoU standard deviation. The interaction between the modules is small and changes sign across seeds, so I describe the effects as complementary and approximately additive. The next question is where this improvement comes from, because the overall score alone does not tell us which cases become easier.

讲述提示：只口述关键结果，不逐格读表。FSDR +0.51；RACE 在 FSDR 上 +0.24；总增益 +0.75。

Thesis source: Tables 5.3–5.4; Figure 5.2

Speaking practice and recall cues

先看图和关键词尝试讲述，再用英文参考段落核对信息。无需逐字记忆。

09 The gains are concentrated in small infections | 9:10–10:50

FSDR precision / RACE recall / smallest quartile plus 1.90 / failures remain

画面建议：仅保留病灶大小四分位的四个增益 1.90、0.68、0.38、0.03 pp；配一个改善例和一个持续失败例。

The error analysis suggests different roles for the two modules. FSDR mainly raises precision and removes false-positive pixels. RACE mainly raises recall and recovers missed foreground. The combination has the highest boundary F1 and the lowest failure rate among the four configurations. More importantly, the gain is concentrated in small infections. In the smallest quarter of test masks, where infection covers less than 4.8 percent of the image, IoU improves by 1.90 points. The patient-level bootstrap interval is from 1.08 to 2.80 points. The gain becomes smaller as lesion size increases and is close to zero in the largest quarter. This makes sense because a relatively small boundary error has a much larger effect on a small lesion. A comparison image can illustrate the recovered region, but the subgroup result is the quantitative evidence. Small, faint lesions also remain a major source of persistent failures, and over-segmentation is still the dominant pixel error. The modules improve a difficult part of the task without resolving it completely.

讲述提示：这是结果解释的核心页。患者 bootstrap 区间放图上；不要把单侧和小病灶当成独立的多份证据。

Thesis source: Tables 5.6–5.7; Figure 5.3; Figure 5.1 or 5.4 for examples

10 What happens when the description changes | 10:50–12:10

Same image and checkpoint / changed text / dependence and limited reorder test

画面建议：只选错配描述和保持区域含义的左右重排两组；明确 baseline 与 full 为三种子。

I also held the images and checkpoints fixed and changed only the descriptions. This experiment uses one image from each of 474 test patients. A mismatched description reduces IoU by about 24 points in both the baseline and the combined model. The networks therefore use the text strongly, but they are vulnerable when that text is wrong. RACE strengthens the connection between the side named in the description and the region that is segmented. A different intervention keeps the same set of regions while changing their left-right order. On this subset, the baseline loses 10.3 IoU points, whereas the combined model changes by only minus 0.3 points. That supports reduced sensitivity to this particular reordering. It does not establish robustness to arbitrary paraphrases. The practical lesson is that stronger use of report guidance must be accompanied by checks on report quality.

讲述提示：错配导致 full IoU 约降 24 pp。保持区域的左右重排：baseline 10.3，full 0.3。只推广到这项具体干预。

Thesis source: Section 5.5.4; Figure 5.8; Table 5.10

Speaking practice and recall cues

先看图和关键词尝试讲述，再用英文参考段落核对信息。无需逐字记忆。

11 External context and computational cost | 12:10–13:10

Close to DD-CMD / different training / parameter cost versus latency

画面建议：精简外部重评表，不混入论文原报分数。底部写 trainable +2.75%、CPU 468 → 865 ms。

Under the shared evaluation pipeline, FReCS-Net is close to the strongest re-evaluated external method. DD-CMD reaches 76.30 percent IoU, compared with 76.23 for my model, although their training recipes and seed replication differ. The matched ablation remains the main evidence for the proposed changes. The combined network adds about 1.07 million trainable parameters, or 2.75 percent over the baseline. However, a small parameter increase does not mean a small runtime increase. In the measured CPU setting, latency rises from 468 to 865 milliseconds, with FSDR responsible for most of the added cost. GPU latency still needs measurement.

讲述提示：承认 DD-CMD 总分略高即可。不要说 lightweight、real-time 或 SOTA。

Thesis source: Table 5.1; Tables 5.11–5.12

12 Limitations and the next evaluation | 13:10–13:50

One benchmark / split and prior Test access / report quality / external validation

画面建议：三条短句：patient-separated validation；incorrect or free-text reports；GPU and efficiency。

The main limits are the data partition, report quality and computational cost. The study uses one benchmark with templated descriptions. Training and validation share 434 recoverable patient identifiers, and the test set had been accessed in earlier project work. A new patient-separated external cohort is therefore the priority. I would also test incomplete and incorrect reports, and measure GPU performance. These studies would establish how far the observed improvements extend beyond the current setting.

讲述提示：将必要限制集中在这一页，用短句说清，不在前面每一页重复。

Thesis source: Sections 6.2.6, 6.3 and 6.4

Speaking practice and recall cues

先看图和关键词尝试讲述，再用英文参考段落核对信息。无需逐字记忆。

13 Conclusion | 13:50–14:20

Selective detail / explicit report agreement / consistent matched improvement

画面建议：两项贡献各一句；76.23% IoU、+0.75 pp、小病灶 +1.90 pp；随后停留并开始问答。

The main conclusion is that selective detail refinement and explicit report-image agreement improve this LViT-based segmentation system. FReCS-Net reaches 76.23 percent test IoU, with a positive improvement in all three paired seeds. The analysis shows complementary error changes and the largest benefit in small infections. Thank you. I welcome your questions.

讲述提示：最后回到两项贡献与证据，不再加新结果。结束后面向评委，按规则进行无电子辅助问答。

Thesis source: Abstract; Section 6.1

Numbers to recall without notes

76.23% IoU；84.71% Dice；总体 +0.75 pp；FSDR +0.51 pp；RACE 加在 FSDR 上 +0.24 pp；最小病灶四分位 +1.90 pp；3 seeds；2,113 Test images；474 Test patients。

A short answer structure

先直接回答，再给一条结果或设计依据，最后补充必要范围。例如：The effects are approximately additive. The interaction averages only plus 0.02 points and changes sign across seeds. I therefore describe complementary contributions, without claiming strong synergy.

问答开始后停止使用练习材料。不要假定可以打开备用页或在屏幕上查数据；若评委另有明确要求，再按其指示操作。

Questions to practise without notes 1–5

目标：每题先用 20–30 秒回答核心；评委追问时再扩展。5 分钟可先按 4–6 个问题模拟，实际数量取决于追问。

01 What is new compared with LViT or existing frequency methods?

创新具体在哪里

FSDR keeps skip, transformer, decoder and text sources separate during frequency-conditioned refinement. RACE adds explicit report-region and visual-evidence agreement before decoding. The contribution is this design and its matched evaluation; I am not claiming that frequency processing or image-text fusion is new by itself.

02 Why use a Haar transform rather than FFT or DCT?

为什么选 Haar

Haar provides a fixed local decomposition with reconstructable low- and high-frequency components, which suits spatial skip refinement. It adds no learned transform parameters. I did not run a controlled Haar-versus-FFT or DCT ablation, so I present suitability as the rationale rather than proven superiority.

03 Does high frequency directly represent lesion boundaries?

高频是否就是病灶边缘

No. High-frequency feature changes include boundaries, normal texture and artefacts. FSDR uses semantic support to regulate their contribution and retains the original skip through a residual path. The design avoids assuming that all sharp changes are useful foreground.

04 Why add two modules instead of one attention layer?

两个模块如何分工

They act at different points. RACE refines encoder skips through report-region agreement, while FSDR controls semantic and detail refinement during decoding. The four-configuration study tests each module with and without the other. Their error profiles suggest complementary precision and recall effects.

05 Is RACE an architectural contribution or just another loss?

RACE 和辅助 loss 的关系

It is an inference architecture with a report prior, visual-evidence branches, regional agreement and gated residual routes. Auxiliary losses give these predictions a defined meaning during training. The three-seed result measures the complete module, including that training, rather than a loss-only effect.

Questions to practise without notes 6–10

目标：每题先用 20–30 秒回答核心；评委追问时再扩展。5 分钟可先按 4–6 个问题模拟，实际数量取决于追问。

06 Does inference require a ground-truth mask or parsed target labels?

推理是否用了 GT

No. Inference uses the image, frozen text features, text attention mask and coordinate basis. Parsed report labels and mask-derived region targets are training supervision. The count outputs share the report head but do not directly enter the spatial routing gate.

07 Are these real clinical reports, and why is text available?

文本来源与实际使用场景

The benchmark provides short descriptions of location and extent at both training and inference. They are structured annotations rather than independently written clinical reports. My evaluation is therefore conditional on having this supplied text; its availability and reliability in a clinical workflow remain to be established.

08 Is an improvement of 0.75 points useful?

平均提升不大如何解释

It is a modest overall improvement, repeated in every paired seed under the shared recipe. It is larger in the smallest lesion quartile, at 1.90 points. Boundary F1 and the failure rate also improve. These are segmentation findings; they do not establish a clinical benefit.

09 What do the standard deviation and confidence intervals mean?

三种子 SD 与患者区间的区别

The standard deviation describes variation across three training seeds. For test-set intervals, I first average per-image differences over seeds, then resample 474 patients with all their images using 2,000 bootstrap replicates. These intervals describe test-sampling uncertainty for the trained models, rather than total training uncertainty.

10 Do the two modules show synergy?

能否说强协同

The effects are approximately additive. The interaction averages about plus 0.02 IoU points and changes sign across the three single-cosine seeds. Both modules help conditionally, but these results do not establish a large or consistently positive superadditive interaction.

Questions to practise without notes 11–15

目标：每题先用 20–30 秒回答核心；评委追问时再扩展。5 分钟可先按 4–6 个问题模拟，实际数量取决于追问。

11 Does the agreement gate protect the model from a wrong report?

错误文本下是否稳健

It does not provide reliable protection in the tested setting. Mismatched descriptions lower the full model's IoU by about 24 points. RACE strengthens the influence of the named side, so incorrect text can move the mask incorrectly. Detecting disagreement and limiting text influence is a priority for future work.

12 Have you shown robustness to wording or paraphrases?

文本不变性结论能推广多远

The evidence covers a specific left-right reordering that keeps the parsed set of regions unchanged. The full model changes by minus 0.3 points, compared with minus 10.3 for the baseline. This does not cover arbitrary paraphrases. Seven unparsed side swaps were grouped with changed meaning; they need separate treatment before broader interpretation.

13 Was the evaluation patient-independent and untouched?

患者重叠与 Test 既往使用

Not fully. Training and validation share 434 recoverable patient identifiers, and the test set had been accessed in earlier development. No recoverable Test overlap was found, but the broader partition limitations remain. The controlled comparisons are informative within this setting; independent patient-separated evaluation is needed next.

14 Why not claim the best performance when DD-CMD is so close?

如何回答外部排名

DD-CMD has the higher measured overall IoU, 76.30 versus 76.23 percent. It also uses a different training recipe and a single retraining seed, while my result averages three seeds. I use this as context and base the architectural claim on the matched PLAM ablation.

15 Is the model lightweight or fast enough for real time?

参数和时延为何不同

The added trainable parameter count is small, about 2.75 percent, but the measured CPU latency increases from 468 to 865 milliseconds. FSDR's memory-bound filtering and decomposition dominate the overhead. I have not measured GPU latency, so I would not claim real-time performance.

Questions to practise without notes 16–20

目标：每题先用 20–30 秒回答核心；评委追问时再扩展。5 分钟可先按 4–6 个问题模拟，实际数量取决于追问。

16 Do the gains survive cosine warm restarts?

为什么主讲只用 single cosine

A separate four-configuration, three-seed warm-restart study also places the combined model first on mean IoU, with a 0.50-point gain over its matched baseline. Individual module effects are less consistent. The single-cosine scores are higher, and I keep the two protocols separate when reporting them.

17 What remains difficult for the model?

失败例告诉我们什么

Small, faint infections remain a major source of persistent failures, and all configurations still have more false-positive than false-negative pixels. Some internal response maps also include borders or markings. The method improves measured segmentation errors but does not provide a lesion-specific explanation at every internal location.

18 How much of the RACE gain comes from routing alone?

完整 RACE 是否能归因到每个分支

The three-seed main study evaluates routing and auxiliary training together. An earlier single-seed control estimated a positive routing increment with auxiliary training fixed, but its image-paired interval crossed zero. Repeated routing-versus-auxiliary controls are needed before assigning a reliable separate gain to routing.

19 Why freeze CXR-BERT, and how are the six regions defined?

冻结文本与区域坐标约定

Freezing CXR-BERT keeps the text representation and training budget shared across the matched experiment. The six regions are image-coordinate halves crossed with three vertical bands; they are not segmented lung lobes. Anatomical interpretation requires compatible orientation conventions because the implementation does not convert patient orientation from metadata.

20 What would you do next, and which limitation is the priority?

下一步研究优先级

First, reserve a new patient-separated external cohort before further tuning. Then test incomplete, incorrect and free-text descriptions, with a mechanism to limit unreliable text guidance. Finally, measure GPU latency and optimise the FSDR implementation. I would repeat the complete four-configuration comparison for these changes.

Rehearsal plan and source map

First pass Understand the story

不计时。只看各页图，回答：问题是什么？我改了哪里？用什么比较支持？为什么小病灶获益？读英文段落是为了确认知识，不是背诵句子。

Second pass Recall from cues

盖住参考段落，用每页 3–4
个关键词说一遍。记录卡住的概念，回到论文补足，再用自己的英语重讲。优先记住 FSDR 与 RACE 的输入、处理、输出。

Third pass Record a timed attempt

以 14:00–14:30 为练习目标，按第一页检查点计时。先完整讲完再回看录音。删去重复定义、逐格读表和实现参数；不要靠提高语速补救。

Fourth pass Simulate the five minute Q and A

关闭所有电子资料，随机抽取问题口头回答。先结论、再依据、再范围。答不出的细节说清楚已做和待验证的内容，不猜测结果。

The day before Check the presentation

PPT 应与这份 13 页结构对齐。新增误差及小病灶页、文本干预页、开销信息；统一项目名称。复核图中字、单位和颜色。必要证据置于主讲，问答不以备用幻灯片为前提。

How to update the October presentation

保留：任务图、LViT 对照、整体结构、FSDR、RACE、匹配四配置和三种子结果。合并：重复结构页、成对消融大表、重复定性整图。新增：小病灶与误差类型、文本干预、参数与实际时延。主讲不展开：完整自适应滤波推导、所有训练运行、探索失败清单、warm-restart 表。它们属于理解准备范围。

Verified sources

Latest thesis: FReCS-Net-thesis.pdf, compiled 28 October 2026; 108 PDF pages. Reviewed 30 October 2026.

Figure 3.1 (printed p.17); Figure 3.2 (p.19); Figures 3.4–3.5 (pp.23–24); Tables 5.3–5.4 (pp.50–51); Tables 5.6–5.7 and Figure 5.3 (pp.53–54); Figure 5.8 and Table 5.10 (p.63); Tables 5.11–5.12 (p.68); Chapter 6. Page numbers here refer to printed thesis pages.

Interpretation note: the seven unparsed side-swap descriptions need an unknown category or manual checking. The talk therefore uses the recognised region-preserving subset and limits the wording claim to that specific intervention.

Thesis SHA-256: 9c6be6116935b52e27a30ca3b62c8b211a7c228ed3b85ece176c95436c5ac680